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CURA

Continuous Welfare Monitoring

Technical Whitepaper

Passive Multi-Modal Wireless Sensing for Individual Vital Sign Monitoring,
Welfare Compliance Logging, and Longitudinal Health Surveillance
in Regulated Care, Custodial and Child Supervision Environments

Version	Classification	Date	Author
1.0	**Confidential**	**May 2026**	**Abhay J Kumar**

Protected under three Australian Provisional Patents

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Local System	Cloud/Federated	Security Hardening

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1. Executive Summary

CURA is a passive welfare monitoring infrastructure platform that continuously monitors vital signs and individual identity of persons in regulated care environments without cameras, wearables, or internet connectivity. It generates legally defensible compliance records for aged care, corrections, childcare, hospital, and disability environments.

$136.5M AUD	122 Patent Claims	10 Environments	0 Cameras
Total Addressable Market	3 Provisionals Filed	Regulated Care Sectors	No Wearables. No Cloud.

Australia's regulated care sector — spanning residential aged care, correctional facilities, childcare services, hospital programs, and disability accommodation — faces a persistent welfare monitoring gap. Manual welfare checks occur every one to two hours, providing momentary observation separated by long unmonitored intervals during which adverse events including falls, breathing cessation, and cardiac events occur undetected. Camera-based monitoring is legally prohibited in sleeping areas. Wearable devices are abandoned by persons with dementia, refused by detained persons, and unsafe for infants.

CURA addresses this gap through passive radio-frequency sensing. WiFi signals already present in care environments are perturbed by the breathing movements and cardiac micro-motion of persons in the monitored space. CURA captures these perturbations using dedicated hardware nodes, processes them through a multi-layer AI pipeline, identifies individual persons without biometric enrolment, and continuously records a comprehensive suite of vital signs per individual. All processing occurs on local hardware within each facility. No data leaves the facility. No internet connection is required.

■ Independent Assessment

CURA is at pre-commercial stage. Performance figures cited in this document derive from published academic research under controlled conditions. Independent clinical validation studies are required before deployment in clinical care pathways. The system is not a medical device and does not diagnose disease. TGA regulatory classification is subject to assessment based on specific deployment configuration.

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2. The Welfare Monitoring Gap

2.1 The Scale of the Problem

Australia's regulated care sector is large, legally complex, and chronically under-monitored. The sector encompasses approximately 3,700 residential aged care facilities housing over 200,000 residents; 200 correctional facilities holding approximately 40,000 persons in custody; 15,000 licensed childcare services enrolling 1.4 million children; several thousand hospital-in-the-home program participants; and tens of thousands of NDIS Supported Independent Living (SIL) residents.

Each sector is subject to specific welfare monitoring obligations under Australian law. The Aged Care Act 2024 requires 24/7 registered nurse presence and continuous duty of care. State and Territory Coroners Acts impose welfare monitoring obligations on correctional services that are subject to coronial inquest when a death in custody occurs. The Education and Care Services National Regulations 2011 require active supervision of children during sleep periods without camera installation in sleep rooms.

2.2 Why Existing Solutions Fail

Approach	Why It Fails	Affected Sectors
Manual welfare checks (every 1-2 hrs)	Snapshot only. Events between checks undetected. Staff workload intensive.	All sectors
Wearable monitors	Removed by dementia residents. Refused by detainees. Unsafe for infants. Compliance < 60%.	Aged care, corrections
Camera-based surveillance	Legally prohibited in bedrooms, cells, and sleep rooms under Australian privacy law.	All sectors
Under-mattress BCG sensors	Single bed only. Cannot identify which person is in the bed. No multi-occupancy capability.	Aged care, hospital
PIR motion sensors	Detects movement only. No vital signs. No individual attribution. Cannot detect breathing.	Aged care, corrections
Cloud-connected platforms	Cannot meet health data sovereignty requirements in corrections and high-security aged care.	Corrections, aged care

2.3 The Regulatory Stakes

The consequences of inadequate welfare monitoring are severe and increasing. Every unwitnessed death in a correctional facility triggers a mandatory coronial inquest. The Royal Commission into Aged Care Quality and Safety (2021) found systemic failures in overnight monitoring. SIDS (Sudden Infant Death Syndrome) remains a significant cause of infant mortality during childcare rest periods. These are not theoretical risks — they are documented, recurring, and legally consequential.

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The specific evidentiary gap is that care facilities cannot currently demonstrate continuous monitoring. A welfare log showing manual checks every two hours does not prove that a resident was alive and breathing at 3:17am. CURA's timestamped, append-only compliance records do.

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3. The CURA Solution

"A manual welfare check every two hours gives you three seconds of observation per resident. CURA gives you continuous monitoring with instant alerts. 65% accurate and continuous beats 100% accurate but two-hourly."

3.1 Core Principle

CURA repurposes WiFi signals already present in care environments as a passive sensing medium. When a person breathes, their chest wall displaces 1-4 centimetres with each breath. This displacement alters the phase and amplitude of WiFi signals passing through the space, creating measurable perturbations across 56 OFDM subcarrier channels. CURA captures these perturbations using ESP32-S3 microcontroller nodes — consumer-grade hardware costing approximately $20 AUD each that exposes the raw channel state information (CSI) hidden by standard consumer WiFi chipsets.

The CSI data is processed through a multi-layer AI pipeline running entirely on local hardware. A dual-branch transformer neural network processes the amplitude and phase streams of the CSI data in separate branches to produce individual body signature embeddings — unique identifiers for each person in the monitored space derived from their body geometry and physiological characteristics, without any biometric enrolment or person cooperation required.

3.2 What Makes CURA Different

Feature	CURA	Existing Solutions
Cameras required	**No**	Often yes
Wearables required	**No**	Usually yes
Internet required	**No**	Most platforms yes
Individual attribution	**Yes — self-supervised**	Rarely
Non-consenting persons	**Yes — no enrolment**	No
Coronial-grade records	**Yes — digital signature chain**	Not specifically
Infrastructure change	**None**	Often significant

4. Technology Architecture

4.1 Seven-Layer Sensing Architecture

CURA integrates up to seven passive sensing layers. Deployments use a subset of layers selected based on environment, budget, and monitoring requirements. Layer 1 (WiFi CSI) is present in all deployments. Additional layers are added for higher accuracy or specific clinical requirements.

Layer	Hardware	Primary Function	Status	Cost (AUD)
L1 — WiFi CSI	ESP32-S3 microcontroller ×6	Breathing, HR, individual ID, presence, falls	Proven — validated in simulation	~$120 per ward
L2 — mmWave Radar	TI IWR6843AOPEVM	HR, HRV, arrhythmia, BP trend, gait	Hardware available — integration pending	~$380 per unit
L3 — Thermal IR	Melexis MLX90640 (32×24px)	Skin temperature, position, person count	Drivers available — trivial integration	~$55 per unit
L4 — BCG Mattress	Piezoelectric film overlay	High-accuracy cardiac waveform, HRV	Commercial products available	~$80-150 per bed
L5 — LiDAR	ToF ranging sensor	3D position, posture, fall geometry	Open source libraries available	~$80-200 per unit
L6 — Passive Acoustic	Microphone array	Cough, wheeze, stridor, respiratory sounds	Local processing only — no audio stored	~$20-50 per unit
L7 — Floor Pressure	Pressure tile array	Gait biometrics, fall impact, exit detection	Custom integration required	~$200-500 per zone

■ Honest Assessment — Layer Maturity

Layer 1 (WiFi CSI) is proven in simulation and the literature. Layers 2-4 have validated hardware and software components but require system integration work. Layers 5-7 are technically sound but not yet integrated into the CURA stack. The Phase 1 pilot product uses Layers 1 and 3. Full seven-layer deployment is a 12-24 month build programme.

4.2 WiFi CSI — How It Works

IEEE 802.11 WiFi operates by transmitting modulated signals across multiple subcarrier frequencies. The Channel State Information (CSI) describes how the signal was modified by the environment between transmitter and receiver — encoding information about every object the signal encountered. Standard consumer WiFi chips discard this information after decoding the data payload. The ESP32-S3 microcontroller

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exposes the raw CSI data across 56 subcarrier channels, providing per-subcarrier amplitude and phase measurements at up to 200 packets per second.

Human breathing displaces the chest wall 1-4 millimetres per breath, shifting the phase of reflected WiFi signals by a measurable fraction of a wavelength. A bandpass filter isolating the 0.1-0.5 Hz frequency range (corresponding to 6-30 breaths per minute) extracts the breathing signal from the CSI amplitude time series. Cardiac micro-motion (approximately 0.1-0.5 mm displacement) produces a weaker signal in the 0.8-2.5 Hz range. Six nodes positioned around a 10-bed ward create 30 crossing signal paths, each providing an independent measurement of the breathing and cardiac signals of each person in the space.

4.3 Processing Stack

Layer	Technology	Function
Firmware	ESP32-S3 + esp-csi (Espressif, MIT)	CSI data acquisition, UDP transmission to processing unit
Sensing server	RuView (Rust, Docker, MIT licensed)	CSI ingestion, vital sign extraction, pose detection
Individual ID	Dual-branch transformer (PyTorch)	Amplitude + phase stream processing, 128-dim body signature
Multi-modal fusion	ResNet + spatial-channel attention	Combining WiFi, radar, thermal, BCG data streams
Vital sign extraction	scipy, BioSPPy, NeuroKit2, librosa	Breathing patterns, HRV, arrhythmia, sleep stage, acoustics
Application API	FastAPI (Python)	REST and WebSocket API for dashboard and integrations
Database	PostgreSQL (append-only schema)	Compliance records with digital signature chains
Staff interface	React	Ward dashboard, floor plan, alert management
Infrastructure	Docker Compose	One-command deployment, watchdog auto-restart
Host hardware	Raspberry Pi 5 (8GB) or Beelink NUC	All computation on-site, no cloud dependency

Entire software stack is open source

All sensing libraries, AI frameworks, and application components are MIT or Apache licensed. Zero licensing fees. Zero cloud subscriptions. CURA's value is in the integration, training pipeline, and compliance architecture — not in proprietary sensing algorithms.

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5. Individual Identification System

The ability to attribute vital signs to specific named individuals is CURA's primary commercial differentiator. Existing WiFi sensing platforms can detect that breathing is present and estimate the count of persons, but cannot determine which vital sign reading belongs to which specific person. CURA's individual identification system is built on published academic research and extends it to address the specific constraints of regulated care environments.

5.1 Dual-Branch Transformer Architecture

The identification mechanism is based on the dual-branch transformer architecture published by Avola et al. (Sapienza University of Rome, IEEE 2023), which demonstrated 99.82% stationary individual identification accuracy using WiFi CSI amplitude and phase streams processed in separate neural network branches on ESP32 hardware. CURA implements this architecture as the identification layer within a regulated care deployment context. CURA's novel contribution is not the architecture itself but its application to non-consenting, non-cooperative persons in regulated care environments for the purpose of vital sign attribution and compliance record generation.

The dual-branch transformer processes CSI amplitude and phase streams in separate branches using multi-head self-attention to extract temporal patterns and cross-subcarrier correlations. The branches are fused to produce a 128-dimensional individual body signature embedding. For stationary persons — sleeping aged care residents, hospital inpatients, detained persons at night — the embedding is highly stable, reflecting consistent body geometry and resting physiological state.

5.2 Self-Supervised Calibration

The calibration process requires no staff action and no person cooperation. The system collects CSI data during a minimum 10-minute rest period, generates embeddings, clusters them using DBSCAN (with person count validated against thermal array data), and assigns persistent identifiers to each cluster. Staff may optionally map identifiers to resident names; the system generates valid compliance records using internal identifiers if names are not provided. This is specifically designed for populations who cannot cooperate with biometric enrolment procedures.

5.3 Identification Accuracy by Environment

Environment	Primary Method	Published Accuracy*	Calibration Time
Aged care beds (overnight)	Stationary WiFi CSI dual-branch transformer	99.82% (stationary)	10 minutes
Childcare sleep room	Stationary WiFi CSI dual-branch transformer	99.82% (stationary)	10 minutes
Corrections single cell	mmWave physiological biometric	96-99%	24-72 hours
Aged care common areas	WhoFi + radar gait fusion	~95% combined	48-72 hours

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Corrections dormitory	WiFi CSI + radar + thermal fusion	>99% multi-modal	72 hours

** Published academic validation under controlled conditions. Independent clinical validation in real care facilities pending. Real-world accuracy will vary.*

■ Critical Limitation

Individual identification accuracy in real multi-occupancy care environments has not yet been independently validated. Published figures derive from controlled laboratory studies with healthy adult volunteers. Aged care residents may have atypical body composition affecting embedding stability. Environmental factors including metal bed frames, air conditioning vents, and thick concrete walls reduce accuracy. The Phase 1 pilot programme will establish real-world baselines.

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6. Individual Vital Sign Suite

CURA monitors a comprehensive suite of vital signs and health parameters per identified individual. Parameters are divided by current technical readiness: those proven in the literature and buildable now; those requiring additional integration work; and those that are longer-term research objectives.

6.1 Phase 1 — Immediately Buildable

Parameter	Source Layer(s)	Alert Threshold	Clinical Relevance
Breathing rate	L1 WiFi CSI	Below 8 or above 30 breaths/min	Respiratory failure, apnoea, SIDS
Breathing presence / apnoea	L1 WiFi CSI	Cessation >15s adults, >10s infants	Emergency alert — cardiac and respiratory
Presence / absence	L1 WiFi CSI + L3 thermal	Absent from expected position	Wandering, fall outside room
Person count	L1 WiFi CSI + L3 thermal	Unexpected count change	Intrusion, absconding
Skin surface temperature	L3 thermal	Deviation from individual baseline	Fever onset, hypothermia
Fall detection	L1 WiFi CSI + L3 thermal	Any detected fall event	Immediate staff response
Body position	L3 thermal + L5 LiDAR	Prolonged same position	Pressure injury risk

6.2 Phase 2 — Integration Required (3-6 months)

Parameter	Source Layer(s)	Clinical Relevance	Build Complexity
Heart rate	L2 radar, L4 BCG, L1 at 5GHz	Tachycardia, bradycardia	Moderate — libraries exist
Heart rate variability (HRV)	L4 BCG + L2 radar	Autonomic state, deterioration predictor	Moderate — NeuroKit2 covers this
Arrhythmia — A-fib, pauses	L4 BCG + L2 radar	Cardiac safety — critical for dementia	Hard — BCG morphology analysis
Sleep stage (N1/N2/N3/REM)	L1 + L2 + L4 fusion	Sleep quality, dementia progression	Hard — needs training data
Breathing pattern classification	L1 + L2	Cheyne-Stokes, apnoea classification	Moderate — signal processing

Gait parameters	L5 LiDAR + L7 floor pressure	Fall risk, neurological deterioration	Moderate — open3d available
Respiratory sounds	L6 acoustic (local processing)	Crackles, wheeze, stridor	Hard — model training needed

6.3 Phase 3 — Research-Grade (12+ months)

Blood pressure trend estimation from pulse transit time (PTT) across BCG and radar modalities; tidal volume estimation from WiFi CSI amplitude modulation depth; HRV DFA alpha-1 for longitudinal deterioration prediction; full polysomnography-equivalent sleep staging with AHI computation; and swallowing detection from passive acoustic signals. These parameters are scientifically supported but require independent clinical validation before deployment.

■ Important Qualification

Blood pressure and tidal volume estimation from passive RF sensing are not clinically validated at the accuracy required for individual patient management. These remain research objectives. CURA does not make clinical diagnostic claims for these parameters.

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7. Deployment Configurations

Residential Aged Care — General Ward

$60-75 per bed per month

Six ESP32-S3 nodes positioned around a 10-bed ward create 30 crossing signal paths. The system monitors breathing, presence, and falls per resident continuously overnight. Individual ID calibrates in 10 minutes. Staff access the dashboard from any LAN browser. Compliance records meet ACQSC quality indicator requirements.

Sensing layers: L1, L3 (Phase 1) + L4, L2 (Phase 2)

Residential Aged Care — Memory Care / Dementia

$75-95 per bed per month

Enhanced deployment with LiDAR and floor pressure arrays for wandering detection and covert exit monitoring. Specifically designed for residents who cannot participate in biometric enrolment and frequently change bed positions. System tracks individual through position changes using multi-modal fusion.

Sensing layers: L1, L3, L5, L7

Correctional Facilities — Single Cell

$70-80 per cell per month

Single mmWave radar unit per cell provides high-confidence vital sign monitoring and physiological biometric individual identification without WiFi infrastructure. Generates continuous per-cell welfare records structured for coronial inquest evidentiary purposes. No cameras. Processing within facility security perimeter.

Sensing layers: L2 (+ L1 where WiFi available)

Childcare / Daycare Sleep Rooms

$50-65 per room per month

No cameras. Per-child breathing monitoring during rest periods. Alert within 10 seconds of breathing cessation. Compliance records satisfy Education and Care Services National Regulations 2011 active supervision requirements. SIDS-pattern breathing detection.

Sensing layers: L1, L3

Hospital-in-the-Home

$85-110 per patient per month

Portable edge node kit for home installation. Full vital sign monitoring without wearable attachment. Consented encrypted vital sign streams to treating clinician dashboard via zero-knowledge relay. HL7 FHIR output for EMR integration.

Sensing layers: L1, L2, L3 portable kit

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Disability Group Homes (NDIS SIL) **$55-70 per resident per month**

Overnight welfare monitoring without staff presence in resident bedrooms. NDIS incident-reporting-compatible event records. Specifically adapted for residents with intellectual disability who cannot maintain wearable devices.

Sensing layers: L1, L3

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8. Intellectual Property — Patent Portfolio

CURA's IP is protected by three Australian provisional patent specifications filed at IP Australia in May 2026. The provisionals lock priority dates for 122 claims across the full CURA architecture. Standard patents must be filed within 12 months (by May 2027). All three provisionals were subjected to multiple rounds of independent technical audit before filing.

AU Provisional AMCZ-2615743059 — **45 claims** · Filed 11 May 2026

System and Method for Passive Multi-Modal Wireless Sensing, Privacy-Preserving Sensor Data Recording, and Comprehensive Individual-Attributed Vital Sign Monitoring, Identity Verification, Welfare Compliance Logging, Clinical Early Warning and Longitudinal Health Surveillance in Regulated Care, Custodial and Child Supervision Environments

Covers: all 7 sensing layers; dual-branch transformer individual ID without biometric enrolment; full vital sign suite per individual (breathing, HR, HRV, arrhythmia, sleep stage, falls, gait, body position, composite health score); privacy-preserving sensor data archival (thermal sequences, LiDAR point clouds, CSI matrices, radar maps, BCG waveforms, acoustic features) per individual; longitudinal health profiles; personalised deterioration prediction; all 10 regulated care environments.

AU Provisional AMCZ-2615744013 — **34 claims** · Filed 11 May 2026

System and Method for Edge-Cloud Partitioned Passive Wireless Vital Sign Monitoring with Heterogeneous-Environment Federated Model Optimisation, Privacy-Preserving Population Health Analytics, and Remote Clinical Oversight in Regulated Care Networks

Covers: CSI feature compression pipeline (93-99% bandwidth reduction); hardware-normalised federated aggregation across heterogeneous WiFi chipsets; layer-adaptive differential privacy for physiological model training; HSM identity partitioning (cryptographic, not administrative); zero-knowledge relay for hospital-in-the-home; population health analytics on de-identified differentially-private aggregates.

AU Provisional AMCZ-2615745256 — **43 claims** · Filed 11 May 2026

System and Method for Security-Hardened Federated Passive Vital Sign Monitoring Networks Comprising Physiologically-Calibrated Byzantine Gradient Aggregation with Dynamic Trust Decay and Server-Side Holdout Validation; Three-Level DP Governance with Multi-Modal Interference Detection; Zero-Gap Short-Lived Certificate Management with Biometric-Wrapped Persistent Storage; Mesh-Extended Dual-Path Alert Delivery with TOTP Device Registration; and Multi-Sensor Quorum-Gated Cryptographic Key Destruction

Covers: Byzantine poisoning detection calibrated to physiological CSI gradient statistics; dynamic trust decay replacing binary probationary graduation; server-side gold-standard holdout validation with permanent node revocation; three-level DP governance (platform ceiling / regulatory floor / operator config); multi-modal interference detection; biometric-wrapped certificate persistence; ESP32 mesh secondary alert network; TOTP device registration; multi-sensor quorum-gated key destruction.

8.1 Prior Art Landscape

A comprehensive prior art search was conducted on 11 May 2026 across Google Patents (including Scholar results), IEEE Xplore, and academic literature. Six targeted patent searches returned zero results for the combination of WiFi CSI sensing with federated learning for physiological monitoring. Two significant academic papers were identified and are documented below with distinction analysis.

Reference	Status	Relevance to CURA
WO2017/156492 (Origin Wireless Inc., 2017)	CEASED — no AU national phase	WiFi CSI breathing detection — basic method is public domain. Does NOT cover individual attribution, vital sign archival, compliance records, or any CURA-specific claims.
AU2023202326 (Hangzhou Normal University)	GRANTED	BCG under-mattress vital signs only. Not WiFi CSI. No individual identification. Not relevant to CURA WiFi claims.
US12279190 (2024)	Granted — US only, no AU filing	CSI breathing with AGC compensation. US jurisdiction only. Not prior art in Australia.
IEEE 10235745 — Avola et al., Sapienza University of Rome (2023)	Academic — not patented	IMPORTANT: Dual-branch transformer processing CSI amplitude and phase separately — achieves 99.82% stationary identification on ESP32. This is the academic source of CURA's 99.82% figure and identification architecture. CURA cannot claim this mechanism broadly. CURA's novel ground: self-supervised calibration for non-consenting persons, vital sign attribution per identified individual, regulated care compliance records — none in this paper.
SADU — Dave, Patel, Sun (TechRxiv preprint, Sep 2025)	Academic — not patented	Multi-user WiFi CSI identification using deep unfolding + manifold attention. 95.9-99.9% identity accuracy. Not patented. Does not cover federated learning, vital signs, or care environments. Requires careful claim drafting at standard patent conversion to distinguish architecturally.

■ Patent Attorney Briefing — Standard Conversion May 2027

At standard patent conversion, the patent attorney must be briefed on IEEE 10235745 (Sapienza 2023). Independent claims must NOT read as "dual-branch transformer processing amplitude and phase separately" — that architecture belongs to Sapienza. Claims must be anchored to the novel application context: self-supervised calibration for non-consenting persons (dementia, infants, detainees), vital sign attribution per identified individual, and regulated care compliance record generation. These applications do not appear in any identified prior art.

Patent Database Search Result — 11 May 2026

Six targeted Google Patents searches combining WiFi CSI + federated learning + vital signs + individual identification returned zero results. The 323-patent broad search was entirely wireless communications patents (Qualcomm, Huawei, LG, Intel) — none in physiological sensing. The patent field is clear. Academic literature requires careful claim drafting around Sapienza 2023.

9. Competitive Landscape

The passive vital sign monitoring space is early but accelerating. Two direct competitors are active in the Australian market. Multiple incumbent technology vendors address adjacent requirements. CURA's competitive position is differentiated by its air-isolated architecture and individual attribution capability.

9.1 Direct Competitors

Talius Group (ASX: TAL) + Inturai

Talius Group is the most significant Australian competitive threat. Talius aggregates sensors for aged care — their clients include Bolton Clarke, Whiddon, Uniting, Anglicare, and Keyton. In January 2026, Talius signed a non-binding LOI with Inturai Ventures (CSE: URAI) for WiFi spatial sensing integration — a $2.5M USD target over three years. Inturai uses standard WiFi signals for health score, sleep monitoring, and presence detection.

Key difference: Inturai/Talius is cloud-based. CURA is local/network-isolated. This is a genuine differentiator for corrections (data sovereignty requirement), high-security aged care, and any facility with strict privacy governance requirements. The cloud architecture also creates per-facility ongoing cloud costs that CURA's local model avoids.

Talius is also approaching Inturai for spatial intelligence but their current product portfolio (PIR sensors, emergency pendants, BCG mattress sensors, ECG smartwatches) does not include passive individual vital sign attribution from WiFi. Their "Umbrella" wearable and "Ikarya" WiFi product are both "Coming Soon" as of the date of this document.

CSIRO / HSC Technology

CSIRO's Smarter Safer Homes platform uses PIR and motion sensors (not WiFi CSI) for behavioural monitoring in residential aged care. Commercialised through HSC Technology (now Talius). Active ARIIA-funded trials with Whiddon. Motion-sensor based — no vital signs, no individual attribution.

9.2 Incumbent Technology Vendors

Vendor	Technology	CURA Advantage
Tunstall / Alarms	Emergency pendants, PIR	Passive — no wearable, no activation required
CardiacSense (via Talius)	TGA-approved ECG+PPG watch	No wearable compliance required for dementia/corrections
Foresite / FallCall	Fall detection sensors	Breathing + vitals, not just falls; individual attribution
Sleep Sense (BCG mattress)	WiFi-connected BCG	Multi-bed, individual attribution; network-isolated
Philips HealthSuite	Cloud health data aggregation	Local processing; no cloud dependency; corrections-ready

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10. Critical Assessment — Limitations and Risks

This section presents an honest assessment of CURA's limitations, risks, and unresolved challenges. It is not marketing material. Any facility, investor, or regulatory body evaluating CURA should read this section.

10.1 Technical Limitations

Individual ID accuracy in real care environments is unproven

Published accuracy figures (99.82% stationary WiFi CSI) derive from controlled laboratory studies with healthy adult volunteers. Aged care residents with atypical body composition, dementia-related movement patterns, or medical devices may produce less stable embeddings. Real-world accuracy requires independent validation in actual care facilities. **Mitigation:** Phase 1 pilot will establish baseline accuracy data. Claims will be qualified as "target accuracy under controlled conditions" until independently validated.

Blood pressure and tidal volume estimation are research-grade, not clinical-grade

Passive PTT-based blood pressure estimation and WiFi CSI tidal volume estimation are theoretically sound but have not been demonstrated at clinically useful accuracy in shared-room passive sensing environments. These remain research objectives, not deployable features. **Mitigation:** Phase 1 does not include these parameters. They are in the patent to stake IP territory for future development.

Multi-person cardiac monitoring accuracy degrades with room occupancy

In a 10-person ward, WiFi CSI heart rate attribution accuracy is estimated at 55-65% compared with 90%+ for breathing. Individual cardiac monitoring requires the mmWave radar layer (Layer 2) for acceptable accuracy. The Phase 1 product (WiFi CSI + thermal only) should not be marketed for cardiac monitoring in multi-occupancy environments.

Environmental factors degrade sensing performance

Metal bed frames scatter WiFi signals unpredictably. Air conditioning vents create periodic motion artifacts. Thick reinforced concrete walls attenuate WiFi signals between rooms. Each facility deployment requires calibration assessment. Some environments may not be suitable for WiFi CSI sensing.

The 7-layer stack is a multi-year build programme, not a current product

The full CURA architecture — all 7 sensing layers, all 19 vital sign parameters, all 10 deployment configurations — represents a 3-5 year development roadmap. The Phase 1 product delivers Layers 1 and 3 with breathing, presence, falls, and basic individual ID. Investors and partners should evaluate CURA against Phase 1 deliverables, not the full patent specification.

10.2 Regulatory Risks

TGA classification: If CURA makes clinical outputs — arrhythmia alerts, deterioration predictions, breathing cessation alerts — it may be classified as Software as a Medical Device (SaMD) under the Therapeutic Goods Act 1989. SaMD Class IIa or higher requires ARTG listing before commercial sale to hospitals or aged care facilities making clinical decisions based on the system output. A TGA regulatory consultant opinion is required before any clinical marketing claims are made. The intended use statement in the patent specification positions CURA as welfare monitoring infrastructure, not a medical device — but specific deployment configurations may cross the SaMD threshold.

Privacy Act 1988: The system identifies individuals and archives per-individual health data. This is personal information under the Privacy Act regardless of the sensing modality. A Privacy Impact Assessment is required before handling real resident data. The local processing architecture substantially reduces privacy risk but does not eliminate compliance obligations.

Coronial evidence admissibility: CURA generates compliance records claimed to be suitable for coronial proceedings. Actual admissibility in specific proceedings is determined by the relevant coroner, not by system design. Chain of custody, calibration records, and system validation documentation will be required to establish evidential weight.

10.3 Commercial Risks

Talius/Inturai market timing: Talius is actively signing MSAs with major aged care operators (Adventist Retirement Plus signed April 2026). If Talius completes Inturai integration before CURA reaches market, the largest aged care group operators may be locked in. CURA's differentiation (local processing, corrections-ready, individual attribution) protects it from full displacement, but speed to market matters.

Alien technology adoption risk: Aged care facility managers and corrections administrators are not technology adopters. Demonstrating that WiFi signals can monitor breathing without cameras or wearables requires significant education. The first pilot will be the hardest sale.

11. Commercial Model and Market Opportunity

11.1 Product Tiers

Tier	Hardware	Identification	Best For	Monthly Fee
CURA Standard	ESP32-S3 nodes ×6	Fixed position assignment	Sleep rooms, single cells, budget	$60 per position
CURA Identify	ESP32-S3 + MLX90640	WiFi CSI dual-branch transformer	Aged care overnight, childcare	$75 per position
CURA Precision	ESP32-S3 + IWR6843 + MLX90640	>95% static + moving	Corrections, aged care common areas	$95 per position

11.2 Total Addressable Market

Sector	Units	Price/Unit/Month	Annual Revenue (10% penetration)
Residential aged care	200,000 beds	$70	$16.8M ARR
Correctional facilities	40,000 cells	$75	$3.6M ARR
Childcare sleep rooms	150,000 rooms	$55	$9.9M ARR
Hospital-in-the-home	50,000 patients	$90	$5.4M ARR
NDIS SIL group homes	30,000 residents	$65	$2.3M ARR
Mental health step-down	15,000 beds	$70	$1.3M ARR
Total TAM (10% penetration)			**~$39.3M ARR**

11.3 Revenue Model and Unit Economics

CURA is sold as a managed hardware-as-a-service subscription. Health Plus International supplies and installs the hardware, provides the software platform, and charges a monthly fee per monitored position. The facility operator has no capital expenditure.

Item	Cost
Hardware per 10-bed ward (6× ESP32-S3 + 2× thermal)	~$230 AUD
Processing unit (Beelink NUC)	~$200 AUD
Installation and configuration	~$300 AUD (one-off)
Total hardware + setup per 10-bed ward	~$730 AUD

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Monthly revenue at $70/bed × 10 beds	$700/month
Hardware recovery period	< 2 months
Gross margin post hardware recovery	>85%
Year 1 target: 10 facilities × 30 positions	$252,000 ARR
Year 2 target: + 5 corrections × 150 cells	$630,000 additional ARR

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12. Implementation Roadmap

Phase	Timeline	Deliverable	Milestone
Phase 0 — IP	May 2026 ■	Three provisional patents filed Trademarks pending	AMCZ-2615743059, -44013, -45256
Phase 1 — Build	Boards arrive → 8 weeks	Working prototype: L1+L3 Breathing, presence, falls, basic ID Staff dashboard Compliance records	Demonstrated working system at Carlingford
Phase 2 — Pilot	Week 9 → Month 6	Free 90-day pilot 1 aged care facility Performance data Testimonial	Accuracy validation, regulatory defensibility confirmed
Phase 3 — Commercial	Month 7 → Month 18	First paid deployments Layer 2 (mmWave) integration Sleep stage, arrhythmia Federated learning MVP	10 facilities, $252K ARR target
Phase 4 — Scale	Month 18 → Month 36	Corrections market entry QLD priority (Talius active there) NSW Justice Health HiTH cloud architecture	$882K ARR target
Phase 5 — IP Conversion	May 2027	Standard patents filed All 3 provisionals converted Patent attorney engagement	122 claims examined and defended

12.1 Immediate Priority Actions

This week: File CURA trademark (Class 10 + Class 42) — $500 at ipaustralia.gov.au/trade-marks

Boards arrive: Flash ESP32-S3 firmware, benchmark breathing detection at home, validate CSI data stream

Month 1: Build Phase 1 software stack — FastAPI, PostgreSQL, React dashboard, Docker Compose deployment

Month 2: Identify aged care pilot facility. Health Plus already has relationships in regional NSW. Queensland is priority — Talius is active there.

Before any clinical marketing: TGA regulatory consultant opinion on SaMD classification. Privacy Impact Assessment. Clinical governance framework.

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13. Regulatory Pathway

Requirement	Priority	Status	Action Required
Australian provisional patents (×3)	DONE	Filed ■	Convert to standard by May 2027
CURA trademark — Class 10 + 42	URGENT	Not filed	File this week — $500
TGA SaMD classification assessment	HIGH	Not commenced	Engage regulatory consultant before clinical marketing
Privacy Impact Assessment	HIGH	Not commenced	Required before handling real resident data
Clinical governance framework	HIGH	Not commenced	Required before any facility pilot
ISO 27001 gap assessment	MEDIUM	Not commenced	Required for government procurement
Essential Eight alignment	MEDIUM	Not commenced	Required for NSW Health / Corrections procurement
Independent clinical validation study	MEDIUM	Pending pilot	Phase 2 pilot generates this data
ACQSC product assessment	LOW	Pending validation	After clinical validation complete
IEC 62443 security certification	LOW	Planned	Before corrections market entry

14. Conclusion

CURA addresses a genuine, large, and legally consequential gap in Australia's regulated care sector. The core technology — passive WiFi sensing for breathing detection and individual attribution — is proven in academic research and validated in simulation. The hardware is available, affordable, and orders have been placed. The IP position is established with three provisional patents covering the full architecture.

The honest assessment is that CURA is a strong concept at pre-commercial stage. The Phase 1 product — breathing, presence, falls, individual ID, compliance records — is buildable now and represents a genuine advance over current manual welfare check practices. The broader vital sign suite, federated learning architecture, and cloud platform are multi-year build programmes that require pilot data, clinical validation, and progressive investment.

The competitive window is open but closing. Talius/Inturai is moving. The Queensland aged care market is being actively contested. The first pilot conversation should happen within 60 days of the boards arriving. The first paid deployment should happen within 12 months.

CURA does not need to be a 19-parameter, 7-layer, federated AI platform to win its first contract. It needs to be better than a manual welfare check every two hours. That bar is cleared with Layer 1 alone.

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Appendix A: Signal Processing Pipelines — Phase 3 Research Parameters

This appendix documents the signal processing pipelines for two Phase 3 research-grade parameters: blood pressure trend estimation from pulse transit time (PTT) and tidal volume classification from WiFi CSI amplitude modulation. Both parameters are included in the CURA patent portfolio and represent longer-term research objectives requiring independent clinical validation before deployment. Neither is included in the Phase 1 or Phase 2 product.

■ Research-Grade Parameters

The pipelines described in this appendix target trend monitoring and individual deviation detection — not absolute clinical measurement. Absolute blood pressure measurement and spirometry-grade tidal volume measurement require contact sensors. CURA's contribution is detecting clinically significant changes from individual baselines without physical sensor attachment.

A.1 Blood Pressure Trend Estimation — Pulse Transit Time Method

A.1.1 Physiological Basis

Pulse Transit Time (PTT) is the time interval between the mechanical trigger of cardiac ventricular ejection and the arrival of the resulting pressure wave at a peripheral measurement site. PTT correlates inversely with arterial blood pressure via the Moens-Korteweg equation: higher blood pressure produces greater arterial wall tension, increased pulse wave velocity, and therefore shorter PTT. The relationship is approximately linear within the physiological range and is stable within an individual over hours to days, making PTT suitable for individual trend monitoring after personalised calibration.

A.1.2 Hardware Requirements

Component	Hardware	Signal	Timing Precision
Cardiac trigger	Layer 4 BCG mattress (piezoelectric)	J-wave peak of BCG waveform — mechanical correlate of aortic valve opening	±10-15ms
Peripheral pulse	Layer 2 mmWave radar (TI IWR6843)	Doppler-detected carotid or radial pulse arrival	±15-25ms
Combined PTT precision	L4 + L2 fusion	Quadrature sum of individual timing uncertainties	±18-30ms
Clinical requirement	Contact PPG (for reference)	Gold standard PTT for calibration sessions	±2-5ms

A.1.3 Signal Processing Pipeline

Step 1 — BCG cardiac trigger extraction. The BCG waveform from the piezoelectric mattress overlay is sampled at 100-1000 Hz. A bandpass filter (1-40 Hz) removes DC offset and high-frequency noise. The J-wave peak — the dominant upward deflection corresponding to aortic valve opening and peak aortic blood flow — is detected using a template-matching algorithm trained on the individual's BCG waveform during baseline collection. Inter-beat intervals are validated against the expected heart rate range (30-200 bpm) to reject motion artefacts.

Step 2 — Radar pulse arrival detection. The IWR6843 mmWave radar operates in continuous wave mode targeting the carotid artery region. Range-Doppler processing isolates the velocity signature of the carotid pulse wave at the expected anatomical range. A matched filter tuned to the expected pulse waveform shape detects the arrival time of each cardiac pulse with sub-frame precision. Frame rate is configured at 50-100 Hz to achieve the required temporal resolution.

Step 3 — PTT computation. For each cardiac cycle, PTT is computed as the time interval between the BCG J-wave peak and the radar-detected carotid pulse arrival. PTT values outside three standard deviations from the 60-second rolling mean are rejected as artefacts. A Kalman filter smooths the PTT time series to reduce beat-to-beat variability while preserving trend information.

Step 4 — Individual baseline and trend detection. During an initial 24-72 hour baseline period, a PTT-to-BP calibration model is fitted for the individual using simultaneous reference cuff measurements at regular intervals (minimum 6 reference measurements). The model is a linear regression: $BP = a - b \times PTT$, with coefficients a and b fitted per individual. The model is updated weekly using overnight PTT distributions to account for gradual physiological change.

Step 5 — Alert logic. Sustained PTT reduction of >10% from individual 60-minute baseline triggers a blood pressure trending alert. A 10% PTT reduction corresponds to approximately 15-25 mmHg systolic BP increase based on published PTT-BP relationships. The alert does not display an absolute BP value — it displays a trend indicator (trending up / stable / trending down) and a severity level (watch / alert / urgent) based on the magnitude and duration of PTT deviation.

A.1.4 Current Accuracy Limitations

Passive PTT estimation currently achieves RMSE of 8-12 mmHg in controlled single-subject conditions (published 2023-2025 literature). Multi-person shared environments, motion artefacts from care activities, and individual anatomical variation (carotid depth, mattress coupling variation with position changes) all increase measurement uncertainty. These limitations make absolute BP values clinically unreliable but do not prevent detection of sustained trend changes exceeding 20 mmHg — which represent clinically significant hypertensive episodes or hypotensive deterioration.

Required precision for individual trend monitoring is achievable. Required precision for absolute clinical measurement is not achievable with current passive sensing hardware. CURA's Phase 3 objective is trend monitoring and hypertensive episode detection — not replacing the sphygmomanometer.

Pipeline Stage	Input	Algorithm	Output

BCG trigger	Piezoelectric waveform 100-1000 Hz	Bandpass filter + template matching	J-wave timestamps (ms precision)
Radar pulse	IWR6843 range-Doppler frames 50-100 Hz	Matched filter + peak detection	Carotid pulse arrival timestamps
PTT calculation	Paired cardiac + pulse timestamps	Interval computation + Kalman filter	Smoothed PTT time series (ms)
Calibration	PTT time series + reference cuff BP	Per-individual linear regression	Individual PTT→BP coefficients
Trend detection	PTT time series vs individual baseline	10% deviation threshold + duration filter	Trending alert (watch/alert/urgent)
Alert output	Trend severity + duration	Clinical alert logic	Staff dashboard trend indicator

A.2 Tidal Volume Estimation — WiFi CSI Amplitude Modulation Method

A.2.1 Physiological Basis

During inspiration, the thorax expands outward by 1-6 mm depending on breathing depth. This displacement modulates the amplitude and phase of WiFi signals passing through the chest region. For a single subcarrier, the received signal amplitude varies approximately sinusoidally with chest displacement for displacements small relative to the signal wavelength (2.4 GHz WiFi: λ = 12.5 cm; typical chest displacement: 1-6 mm = 0.8-4.8% of λ). In this regime, CSI amplitude modulation depth is approximately linear with tidal volume for a given individual at a fixed position.

The key challenges are: (1) position dependence — the amplitude-volume relationship changes substantially with the person's orientation relative to the antenna; (2) inter-individual variation — body composition, chest wall compliance, and lung anatomy create significant individual differences; (3) multi-person interference — in shared rooms, each person's breathing modulates signals from all node pairs. These challenges are addressed through individual calibration baselines, multi-node averaging, and independent component analysis for person separation.

A.2.2 Hardware Configuration

Tidal volume estimation uses the Layer 1 WiFi CSI infrastructure already deployed for breathing rate detection. No additional hardware is required. The six ESP32-S3 nodes in a standard ward deployment create 15 independent node pairs, each providing an independent CSI amplitude time series. Multiple node pairs at different angles relative to each person reduce the position dependence of individual measurements.

A.2.3 Signal Processing Pipeline

Step 1 — Per-person CSI extraction. Individual identification from the dual-branch transformer assigns each bed position a persistent identity. For each identified person, the system selects the 3-4 node pairs with the highest signal-to-noise ratio for that person's position, based on the CSI breathing signal amplitude at the

breathing frequency. Subcarriers in the 0.1-0.5 Hz band are isolated using a bandpass filter.

Step 2 — Amplitude envelope extraction. For each selected node pair and subcarrier, the CSI amplitude time series is processed using a Hilbert transform to extract the instantaneous amplitude envelope. The envelope represents the depth of chest displacement modulation on the WiFi signal. Breath-by-breath peak amplitude values are extracted for each respiration cycle using peak detection on the filtered envelope.

Step 3 — Multi-node fusion. Amplitude envelope values from the 3-4 selected node pairs are combined using a weighted average, with weights proportional to the signal-to-noise ratio of each node pair for that person. This averaging reduces position-dependent measurement variation and suppresses interference from other persons' breathing signals via spatial diversity.

Step 4 — Individual calibration and classification. During baseline collection, the individual breathes at rest for 15-30 minutes, establishing a reference amplitude distribution corresponding to their normal tidal volume at rest (~500 mL). Rather than estimating absolute tidal volume in mL, the system classifies breathing depth into four categories relative to individual baseline: shallow (<60% baseline amplitude), reduced (60-85%), normal (85-115%), and deep (>115%). This categorical approach avoids the position-dependence problem of absolute mL estimation.

Step 5 — Alert logic. Sustained shallow breathing classification (>60 seconds at <60% baseline amplitude) triggers a reduced tidal volume alert. This threshold corresponds to clinically significant hypoventilation — breathing depth insufficient for adequate gas exchange at normal respiratory rate. The alert is combined with breathing rate data: shallow breathing at normal rate suggests a different clinical picture than shallow breathing at elevated rate (tachypnoea).

A.2.4 Current Accuracy Limitations

Published WiFi CSI tidal volume estimation achieves approximately 15% RMSE in single-subject controlled-position conditions (CMU 2024, ETH Zurich 2024). In free-position, multi-occupancy conditions this degrades to 25-35% RMSE for absolute values. Categorical classification (shallow/normal/deep) is more robust — estimated classification accuracy of 80-85% in multi-occupancy conditions based on published signal separation performance.

The 15% RMSE limitation means absolute tidal volume values (mL) are clinically unreliable. However, detecting a sustained reduction from 500 mL to below 300 mL — the clinically significant hypoventilation threshold — requires only detecting a >40% amplitude reduction, which is well above the measurement noise floor. CURA's Phase 3 objective is hypoventilation detection and breathing depth trend monitoring, not spirometry.

Pipeline Stage	Input	Algorithm	Output
Person isolation	CSI from 6 ESP32-S3 nodes	Dual-branch transformer ID + node pair selection	3-4 best node pairs per person
Envelope extraction	CSI amplitude time series	Bandpass 0.1-0.5 Hz + Hilbert transform	Breath-by-breath amplitude envelope

Multi-node fusion	Envelopes from 3-4 node pairs	SNR-weighted average	Fused amplitude per breath
Calibration	Resting amplitude distribution	Individual baseline percentile model	Normal amplitude reference
Classification	Fused amplitude vs baseline	Threshold classifier (60%, 85%, 115%)	Depth class: shallow/reduced/normal/deep
Alert output	Sustained shallow classification	>60s threshold + rate combination	Hypoventilation alert with severity

A.3 Combined Clinical Picture — Deterioration Pattern Detection

The clinical value of combining PTT-based BP trend monitoring with tidal volume classification is greater than either parameter alone. Specific deterioration patterns produce characteristic multi-parameter signatures detectable by CURA before they become acute emergencies:

Clinical Event	Breathing Rate	Breathing Depth	BP Trend	HR	Pattern Onset
Respiratory failure (early)	Elevated (>20)	Reduced → shallow	Trending up	Elevated	1-4 hours before acute
Hypertensive crisis	Normal or elevated	Normal	Trending up rapidly	Elevated	30-90 min before symptoms
Sepsis (early)	Elevated	Shallow	Trending down	Elevated	2-6 hours before acute
Cardiac decompensation	Elevated	Shallow (pulmonary oedema)	Variable	Elevated/irregular	1-8 hours before acute
Opioid overdose	Reduced (<10)	Very shallow	Trending down	Reduced	Minutes
Hypoglycaemia	Normal	Normal	Trending down	Elevated	15-60 min before symptoms

Pattern-based deterioration detection using the composite of all available CURA parameters is more sensitive and specific than any individual threshold alert. The Composite Individual Health Score (CIHS) described in the main patent specification implements this multi-parameter pattern matching, with individual baselines providing the personalised reference against which deterioration is measured.

Appendix A documents research-stage signal processing pipelines. Neither pipeline is validated for clinical use. Independent clinical validation studies are required before these parameters are used in patient care decisions. Patent references: AMCZ-2615743059 (PTT blood pressure trend, tidal volume estimation). © 2026 Westminster Green Solutions T/A Health Plus International.